**ROLL NO:** _____

# MADHAV INSTITUTE OF TECHNOLOGY & SCIENCE, GWALIOR

(A Deemed to be University | Autonomous Institute Established in 1957 | NAAC 'A++' Accredited)

## END-SEM EXAMINATION â€" DECEMBER / JUNE 2024

FIRST YEAR (COMMON TO ALL BRANCHES) â€" SEMESTER 1

**Time Allowed: 3 Hours** **Maximum Marks: 70**

*NOTE: 1. Attempt any FIVE questions selecting at least ONE question from each Unit.*
*2. All questions carry equal marks. Sub-questions carry marks indicated in brackets [ ].*
*3. Assume suitable missing data, if any, and state it clearly.*
*4. Use of scientific non-programmable calculator is permitted. Mobile phones are strictly prohibited.*

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### --- UNIT I (Wave Optics) ---

**Q.1** (a) Explain the phenomenon of interference in thin films due to reflected light. Derive the condition for constructive and destructive interference. [7 Marks]
(b) Describe Newton's Rings experiment. How can you determine the wavelength of sodium light using Newton's rings? [7 Marks]

**[ OR ]**

**Q.1 (Alt)** Alternative: (a) Differentiate between Fresnel and Fraunhofer diffraction. Explain Fraunhofer diffraction at a single slit. [7 Marks]
(b) What is Resolving Power of a grating? Derive Rayleigh's criterion for resolution. [7 Marks]

### --- UNIT II (Lasers & Fiber Optics) ---

**Q.2** (a) Explain Einstein's A and B coefficients. Derive the relation between spontaneous emission and stimulated emission. [7 Marks]
(b) Describe the construction, working, and energy level diagram of Ruby Laser and He-Ne Laser. [7 Marks]

**[ OR ]**

**Q.2 (Alt)** Alternative: (a) Explain the principle of optical fiber communication. Derive expressions for Acceptance Angle and Numerical Aperture (NA). [7 Marks]
(b) Compare Step-Index and Graded-Index optical fibers. Discuss attenuation and dispersion losses. [7 Marks]

### --- UNIT III (Quantum Mechanics) ---

**Q.3** (a) Describe de Broglie's hypothesis of matter waves. Explain the Davisson-Germer experiment and its significance. [7 Marks]
(b) State Heisenberg's Uncertainty Principle. Prove that an electron cannot reside inside an atomic nucleus. [7 Marks]

**[ OR ]**

**Q.3 (Alt)** Alternative: (a) Derive time-independent Schrödinger wave equation for a free particle. [7 Marks]
(b) Apply Schrödinger equation to calculate energy eigenvalues and wavefunctions of a particle confined in a 1D infinite potential well. [7 Marks]

### --- UNIT IV (Solid State Physics & Superconductivity) ---

**Q.4** (a) What are Miller Indices? Explain the procedure to find Miller Indices of crystal planes. [7 Marks]
(b) State Bragg's Law of X-ray diffraction. Explain the working of Powder X-ray Diffractometer. [7 Marks]

**[ OR ]**

**Q.4 (Alt)** Alternative: (a) Explain Superconductivity, Critical Temperature, and Critical Magnetic Field. [7 Marks]
(b) Describe the Meissner Effect. Differentiate between Type-I and Type-II superconductors. [7 Marks]

### --- UNIT V (Electromagnetism & Nanotechnology) ---

**Q.5** (a) State and explain Maxwell's four equations in differential and integral forms. [7 Marks]
(b) What is Poynting Vector? State and prove Poynting's Theorem of energy conservation in electromagnetic fields. [7 Marks]

**[ OR ]**

**Q.5 (Alt)** Alternative: (a) What is Quantum Confinement? Explain Quantum Wells, Quantum Wires, and Quantum Dots. [7 Marks]
(b) Describe the Sol-Gel and Chemical Vapor Deposition (CVD) methods for synthesizing nanomaterials. [7 Marks]

**** END OF QUESTION PAPER â€" MITS GWALIOR AUTONOMOUS EXAMINATION SYSTEM ****